

Adnan Ahmad

AI / Machine Learning Engineer • Computer Vision • NLP / LLMs

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SUMMARY

Machine Learning Engineer with a PhD in ML and 13+ peer-reviewed publications, specializing in building and deploying AI systems across **computer vision, NLP/LLMs, and multimodal deep learning**. Hands-on experience shipping real-time inference pipelines (PyTorch, FastAPI, Docker, GCP), retrieval-augmented generation (RAG) systems, and large-scale distributed training. Seeking a Computer Vision / NLP / AI Engineer role turning research-grade models into reliable products.

TECHNICAL SKILLS

Languages: Python, SQL, C++, Java

ML & Deep Learning: PyTorch, TensorFlow, Scikit-learn, CNNs, Transformers, Representation / Continual / Self-Supervised Learning, Reinforcement Learning

NLP & LLMs: Hugging Face Transformers, RAG, LangChain, SentenceTransformers, LLM Fine-Tuning (LoRA / PEFT), Prompt Engineering, Vector Databases (Qdrant)

Computer Vision: OpenCV, Image Classification & Detection, Multimodal (image / signal) Modeling, Medical Image Analysis

MLOps & Cloud: Docker, FastAPI, Git, MLflow, Weights & Biases, CUDA, Google Cloud Platform (Vertex AI, Compute Engine)

Data: Pandas, NumPy, Data Pipelines, Distributed / Federated Training, Real-Time Streaming (LSL)

EXPERIENCE

AI / Machine Learning Scientist

March 2024 – May 2025

Deakin University — Applied ML & Human-AI Systems Lab

Waurin Ponds, VIC, Australia

- Built and deployed **real-time multimodal ML systems** that process live physiological signal streams (EEG/ECG/eye-tracking) through low-latency inference pipelines, served as containerized microservices on GCP (Docker, FastAPI).
- Developed **continual-learning** models for non-stationary data streams, improving cross-subject generalization and reducing accuracy degradation under distribution shift.
- Designed **reinforcement-learning** agents for human-AI collaboration using hierarchical skill learning, improving task coordination in both simulation and human-in-the-loop deployments; results published in 4+ papers.

Machine Learning Scientist (Federated & Distributed Learning)

July 2024 – Jan 2025

National Research Council of Italy (IIT-CNR) — ERCIM Fellowship

Pisa, Italy

- Designed **decentralized federated-learning** algorithms for peer-to-peer networks handling data, model, and communication heterogeneity with no central coordinator.
- Built a second-order (Hessian-based) aggregation method in PyTorch that **reduced communication rounds and sped up convergence** versus FedAvg on benchmark vision datasets; work under review at IEEE TETCI.

Machine Learning Teaching Assistant

March 2021 – March 2024

Deakin University

Burwood, VIC, Australia

- Led workshops on deep learning, **computer vision**, and **NLP**, plus core AI (search, optimization, agent-based reasoning) for cohorts of postgraduate students.

PROJECTS

- Document-QA RAG System** — End-to-end retrieval-augmented generation service: ingests PDFs, embeds chunks with SentenceTransformers into a Qdrant vector store, and serves grounded answers from an LLM through a FastAPI + Docker deployment. [Project Page](#) | [GitHub](#)
- Real-Time Human-AI Coordination via Physiological Feedback** — Real-time Overcooked system where models infer human mental state from EEG/ECG/eye-tracking and RL agents adapt in response; built on LSL streaming, Docker, and FastAPI APIs. [Project Page](#) | [GitHub](#)
- Real-Time Conversational AI for Pepper Robot** — Speech-to-response conversational system integrating a SoftBank Pepper robot with GPT via an API-driven, low-latency architecture. [Project Page](#) | [GitHub](#)
- Decentralized Federated Learning (DecFW)** — PyTorch peer-to-peer training framework with a Hessian-based aggregation method that beats FedAvg on convergence and communication cost under non-IID data. [Project Page](#) | [GitHub](#)

EDUCATION

PhD in Machine Learning — Deakin University, Australia

Dec 2020 – Sep 2024

Thesis: Robust and Communication-Efficient Federated Learning under Non-IID Data.

BSc in Electrical (Computer) Engineering — COMSATS University, Pakistan

Sep 2013 – Sep 2017

PUBLICATIONS & AWARDS

13+ peer-reviewed papers in Machine Learning, Pattern Recognition, Neurocomputing, ACM TIST, and PAKDD. Reviewer for NeurIPS, ICLR, and Pattern Recognition. Full list: [Google Scholar](#).

Awards: ERCIM “Alain Bensoussan” Fellowship (2024); Chartered Professional Engineer (Engineers Australia); Deakin Cyber Research Grant (AUD 12K); Postgraduate Research Scholarship (2020–2024).